Trading Activity of Commodity Futures and Options Around USDA Announcements

Tianyang Zhang

Abstract

The USDA announcement effects on commodity options have received limited attention in the literature. We find that the trading volumes of futures and options fluctuate significantly before and after the release of the monthly World Agricultural Supply and Demand Estimates (WASDE) reports for seven agricultural commodities. The effects of the announcements on commodity options are less significant than those on futures. Furthermore, we provide evidence that informed trading exists in commodity markets before the release date. The information contained in the commodity futures and options has a longer predictive horizon for post-announcement futures returns than the ending stocks forecasts in the reports.

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1 Introduction

As a major public source of forecasts since the year 1973, the World Agricultural Supply and Demand Estimates (WASDE) reports from the U.S. Department of Agriculture (USDA) play an important role in agriculture markets. Not surprisingly, the announcement effects of USDA predictions on agricultural markets have been studied a lot. Most of these studies focus on how the commodity futures markets are affected and provide evidence to confirm whether the effects exist. For example, Adjemian and Irwin (2018) finds that the volatility of agriculture futures spikes right after the release of WASDE reports but declines in a few minutes. Adjemian (2012) analyzes the absolute value of overnight return before and after the announcement date and confirms that the USDA announcement effect persists across contract positions. There is also evidence that the co-movement of different commodity futures returns exists around the release date of USDA predictions (Karali, 2012). Other prominent examples of this line of work include Irwin et al. (2001), Indriawan et al. (2021), and Ying et al. (2019), among others.

However, as a critical part of commodity markets, whether commodity options markets are affected by USDA announcements has been ignored in most cases. One example is from Randall and Summer (1993), who use the daily data from year 1968 to 1989 and provide some evidence that there is a USDA report effect in commodity futures and options markets. In recent decades, the commodity options markets have developed rapidly and become an important derivative in commodity markets. A commodity option contract is written with a particular futures contract as the underlying security. Buying a call (put) option gives the owner the right to buy (sell) the underlying commodity futures at a fixed price or the strike price on or before the contract expiration date. The physical commodity option was first introduced on February 15, 1985, as the Commodity Futures Trading Commission (CFTC) approved Amex Commodity Corporation's physical gold bullion option contract.

As an emerging market, how the commodity options market works has not been studied much in the previous literature. Recently, Zhang (2022) examines hedging pressure and

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liquidity provision in commodity options markets. Cheng et al. (2021) also confirms the existence of hedging premiums and demand effects in commodity options markets. In addition, empirical results find that the commodity options can have a large effect on the underlying asset commodity futures (Luo et al., 2021).

The present study focuses on the commodity options for two main reasons. First, producers would employ options to hedge production and other risks. Lapan et al. (1991) build a theoretical model and shows that when market prices are biased, commodity options must be included to construct optimal hedging strategies for the producers. Moschini and Lapan (1995) also shows that the options can be used to hedge the joint production and price risk for production firms. Therefore, after the WASDE reports are released every month, producers have an incentive to adjust their hedging strategies by trading commodity futures and options. This will cause changes in trading behaviors of options and futures after the announcements. For example, Cao and Robe (2022) find that the corn and soybean option-implied volatilities are significantly lower for several trading days after the reports are released. In this study, we provide new evidence that there exists USDA announcement effects in commodity options markets, although these effects are less significant than those in commodity futures markets.

Second, commodity traders would use options to speculate and make profits. From the perspective view of informed trading, the trading activity in options markets also contains information about the future condition of agriculture markets. Empirical analysis shows that options lead futures in reflecting new information in most commodities and speculation is a key determinant of price discovery (Bohmann et al., 2019). Thus, an interesting topic is to explore and compare the predictability of information in commodity markets and WASDE reports before the announcements. In this study, we use ending stocks forecasts in the reports as benchmark forecasts.

Our study is also related to the literature on the forecasts from financial markets for agriculture information. This study not only investigates the USDA announcement effects in

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commodity futures and options markets but also explores the information content in these two markets in the pre-announcement period. Informed market participants may possess information pertaining to WASDE reports and commodities markets that extends beyond the information contained within the reports. We found that the information in commodity options and futures markets has a longer predictive horizon than the ending stocks forecasts from WASDE reports.

We make three main contributions to the literature. First, we investigate the USDA announcement effects on commodity options markets based on trading volumes, which has not been empirically studied in the previous literature. Second, our study finds that the USDA announcement effects in commodity futures are more significant than that in commodity options, while the options markets could be involved with more informed trading activities before the WASDE reports are released. Third, the ending stocks forecasts are observed to exhibit predictive efficacy within the five trading days following the release, with this predictive capability diminishing rapidly thereafter. More importantly, this study concludes that the information in commodity futures and options markets has longer predictive power than the ending stocks forecasts in the WASDE reports.

The remainder of this paper is organized as follows. Section 2 describes the data of commodity markets and WASDE reports used in this study. The empirical analysis of USDA announcement effects on commodity futures and options markets is in Section 3. Section 4 presents the predictive power of trading volumes in commodity markets and the ending stocks forecasts from the reports. Section 5 concludes.

2 Data

The data of commodity markets used in this study are from Bloomberg and comprise individual futures contracts for seven agricultural commodities including soybean oil, corn, cotton, soybeans, oats, sugar, soybean meal, and wheat. This study encompasses data span-

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ning from January 1995 to December 2018. We follow Zhang (2022) and select a sample of each commodity according to the options expiration dates, which is presented in Figure 1. Specifically, for each futures contract, we include the sample from the Tuesday of the week before the option expiration date to 65 calendar days before the expiration date of the option. To avoid the problems of outliers and the low liquidity in commodity futures and options markets, we drop the two-sided 1% data based on the value of options to futures trading volumes ratio. By applying the specified data constraints, the sample in our study comprises 42691 commodity-daily observations.

The summary statistics for commodities are in Table 1. The trading volumes of futures and options are larger in corn markets than the other commodities, while cotton and soybean meal have the smallest volumes in futures and options, respectively. Soybeans and wheat hold the second and third positions in terms of both futures and options trading volumes. There is an observable phenomenon that a commodity with substantial futures trading volumes typically exhibit corresponding high options trading volumes. However, this trend does not hold true for cotton, as it possesses the lowest futures trading volumes, yet its options trading volumes rank as the third lowest among all seven commodities. Thus, it is not surprising that the options-to-futures trading volumes ratio is largest in cotton, while the lowest ratio is observed in the soybean oil markets.

The forecasts of commodity markets are from USDA, an entity that publishes the monthly WASDE reports to announce current and expected market conditions for agricultural commodities to participants in commodity markets. The number of USDA monthly forecasts including in this study is 288, since the the data covers the period from January 1995 to December 2018. A crucial forecast from WASDE reports is the expected ending stocks at the end of each marketing year. Ending stocks, also referred to as carryout, are the amount of a commodity left over after all demand has been satisfied and enters the supply side of the market in the following marketing year. According to Good and Irwin (2014), ending stocks may be the single most important factor that summarizes the price implications for

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a commodity. Therefore, in this study, we set the ending stock forecast as the target to compare the predictive power between the information in commodity markets and WASDE reports. To the best of our knowledge, this study is the first to investigate the predictive capacity of ending stocks forecasts in futures returns of agriculture commodities.

3 USDA announcement effect

3.1 Abnormal values of futures and options volumes

Intuitively, the futures and options trading activities around the announcement date could be significantly different from the activities on other dates in the same month. In order to investigate the announcement effect, we conduct the analysis about whether there are significant abnormal values of futures and options trading volumes, just before and after the announcement date.

Following Adjemian (2012), an event window of eleven days, is established before and after report publication. First, we separate the trading days for commodity i in month m into two groups according to the announcement date W_m. One group consists of the ordinary trading days $\mathbf{W}_m$ that are in the interval $[W_m - 5, W_m + 5]$, which covers the five trading days before and after the announcement. Another group is made of the trading days $\mathbf{W}_{-m}$ that are outside of the days in interval $[W_m - 5, W_m + 5]$ in the same month. Next, for commodity i in month m on day t, the measures for abnormal values of futures and options volumes are constructed as the following:

$$ab_FVOL_{i,m,t} = |FVOL_{i,m,t} - mean(FVOL_{i,\mathbf{W}_{-m}})|$$

$$ab_OVOL_{i,m,t} = |OVOL_{i,m,t} - mean(OVOL_{i,\mathbf{W}_{-m}})|$$

Table 2 shows the abnormal trading volumes in futures and options during the five trading days both preceding and following the announcement date (Dw) for each commodity. There

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exists a relatively stable pattern of abnormal futures trading volumes during the five trading days leading up to the release date (from Dw-5 to Dw-1) for all commodities. However, it is evident that abnormal futures volumes on the announcement date exhibit the highest values among all eleven trading days (from Dw-5 to Dw+5), across all commodities. A declining trend in abnormal futures volumes is observed following the announcement as the abnormal values in Dw+5 are lowest in the sample except for wheat and sugar. With respect to abnormal options volumes, the values in Dw remain the highest, with the exception of sugar. Different from the futures markets, abnormal options volumes exhibit higher volatility prior to the announcement. Following the announcement, there is a decrease in abnormal options volumes except for sugar, which is similar to the observed patterns in abnormal futures volumes. An interesting finding is that, on the announcement date, the abnormal options volumes of sugar are lower compared to those observed in Dw-5 and all five trading days following the announcement.

In ordr to estimate the USDA reports effects on commodity markets, we conduct an empirical analysis similar to Adjemian (2012). $[W_m - 5, W_m + 5]$ are the set of trading days in the event window when the report is published in month m on day W_m. The dummy variable vector D_{W_m} equal to one if day t is the date W_m and zero otherwise:

$$D_{W_m,t} = \begin{cases} 1 & \text{if } t \in W_m \\ 0 & \text{otherwise} \end{cases}$$

Next, regarding the five days before and after the date W_m, we construct the other dummy variable vectors as:

$$D_{W_m-j,t} = \begin{cases} 1 & \text{if } t \in W_m - j \\ 0 & \text{otherwise} \end{cases}, \text{for j} = 1,2,...,5$$

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$$D_{W_m+n,t} = \begin{cases} 1 & \text{if } t \in W_m + n \\ 0 & \text{otherwise} \end{cases}, \text{for n = 1,2,...,5}$$

Let $\mathbf{DUM}_{Event}$ indicate the dummy variables matrix that makes up the event window of $[W_m - 5, W_m + 5]$. To account for the seasonality and time effects, we also add the dummy variables for months ($\mathbf{DUM}_{month}$) and time trend ($\mathbf{DUM}_{trend}$). We construct the dummy matrix $\mathbf{DUM}$, including $\mathbf{DUM}_{month}$ and $\mathbf{DUM}_{trend}$. The matrix of the control variables is $\mathbf{X}$, which includes the trading volumes and return in the previous trading day of the announcement date. Thus, the cross-sectional regression in our study is:

$$\mathbf{y} = \mathbf{DUM}'_{\mathbf{Event}}\beta + \mathbf{X}'\delta + \mathbf{DUM}'\theta + \epsilon$$

where $\mathbf{y}$ are daily abnormal futures or options trading volumes and ϵ represents the error terms. All estimates are obtained using the OLS method, correcting for serial correlation by the Newey-West method.

Tables 3 to 6 and Figures 2 to 5 report the results of cross-sectional regressions and show the abnormal values of futures and options volumes associated with the event window and the marginal effect of the WASDE reports on the announcement day. In the futures markets, there are distinct spikes of abnormal futures volumes on the announcement date across all seven commodities, which indicates that the abnormal values reach their peak on the release date within the event window, which encompasses the five trading days before and after the announcement date. On and after the release date, when the traders gain new information from WASDE reports, whether it is good or bad news, they would adjust their trading positions in accordance with the new information they get. However, the reactions of different commodity markets are not the same. For corn, soybeans, and wheat, there is a sharp increase in abnormal futures volumes on the day of the announcement accompanied

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by a rapid decrease in abnormal values just one day after the announcement, which indicates that the information from WASDE reports is reflected on the futures contracts shortly in these markets. In cases of the other four commodities including soybean oil, cotton, sugar, and soybean meal, the decreasing trend of abnormal futures volumes after the announcement also appears, but the decline rate is lower. Regarding the abnormal futures volumes in the pre-announcement period, there are also relatively significant abnormal futures volumes on days before announcement compared with that on announcement day in soybean oil, cotton, sugar, and soybean meal futures markets. A plausible explanation for this phenomenon lies in traders' capacity to adapt their futures investments in accordance with their anticipations of commodity forecasts preceding the release date of these reports. This would result in greater fluctuations in trading volumes within the five days leading up to the release date compared to the trading volumes observed during the remaining days of the month. Based on the findings above, we can conclude that futures markets would react to the news from WASDE reports right away after the announcement. In most cases, such reactions tend to subside within five days following the release date.

In commodity options markets, the impact of announcements is comparatively less significant than that observed in futures markets. In the cases of corn and soybeans, the behaviors of options are similar to futures, with abnormal options volumes peaking on the release date and subsequently diminishing within five trading days. However, it is important to highlight that in the soybean oil, cotton, sugar, soybean meal, and wheat options markets, there is an absence of distinct spikes in abnormal options volumes on the release date. Following the announcement date, it takes a longer time for options volumes back to normal in comparison to futures volumes across most commodity markets. This suggests that there is a prolonged period required for the news in WASDE reports to be fully incorporated in options markets. Interestingly, the abnormal options volumes are even higher after the announcement date in soybean oil, sugar, and sugar meal markets. One plausible hypothesis is that options traders may require a longer time to execute adjustments to their positions in options as compared

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to futures. During the pre-announcement period, notably substantial abnormal options volumes are observed, particularly in soybean oil, cotton, sugar, and soybean meal. Compared with futures markets, this phenomenon is more notable in options markets. One possible explanation is that there are more informed trading activities involved in options markets and informed traders appear to leverage their private information regarding WASDE reports to adjust their positions in options to hedge risks or pursue profits.

4 Post-announcement returns analysis

In the previous section, results show that significant abnormal futures and options volumes exist in the pre-announcement period, which could be evidence that some market participants make investments based on their private information before the WASDE reports are released. Therefore, in this section, we examine the predictive power of futures and options trading volumes for futures returns following the announcement. We use four return windows: CUM(+0,+5), CUM(+0,+10), CUM(+0,+15), and CUM(+0,+20). CUM(X,Y) indicates the cumulative return for each commodity from X trading days to Y trading days after the announcement date.

In order to compare the predictive capability of futures and options volumes with the forecasts in WASDE reports, we set the ending stocks forecasts as the benchmark target. Ending stocks is the expected inventory of a particular commodity at the end of the marketing year. In every month, WASDE reports provide the prediction of ending stocks for each commodity. The forecast of ending stock for commodity i in month m is defined as $ES_{i,m}$. To capture the new information released at the announcement, the surprise of the forecast in ending stocks in month m for commodity i is constructed as the difference between the forests in this and the previous month divided by the forecast in the previous month:

$$\Delta ES_{i,m} = \frac{ES_{i,m} - ES_{i,m-1}}{ES_{i,m-1}}$$

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The pre-announcement returns may have substantial influence on the post-announcement returns since the informed traders could obtain private information before the announcement and start to trade in the same direction as the reactions from the announcement. Motivated by this rationale, we construct two more variables: pre-CUM and abs(pre-CUM) denote cumulative return and the absolute value of cumulative return over the pre-announcement window (days -5 to -1). Additional control variables include CAR and AMI, which are the average of daily futures basis and illiquidity measure for the commodity over the pre-announcement window (days -5 to -1) [1]. Referring to Petersen (2009), to correct for cross-sectional correlation, the standard errors are clustered (by time). The employed cross-sectional regression is as follows:

$$R_{i,t+1} = a_t + \mathbf{b}_t^{'} \mathbf{X}_{i,t} + e_{i,t+1},$$

where $R_{i,t+1}$ is futures return at post-announcement period $t+1$ for commodity i, $\mathbf{X}_{i,t}$ is a vector of characteristics known as of announcement date, and $e_{i,t+1}$ is a residual.

Table 7 presents the results about the predictive power of futures, options volumes, and ending stock forecasts on the cumulative returns following the announcement. In the column of CUM(+0,+5), the coefficient on ΔES is significantly negative (t-statstic = -3.759). The forecast of higher ending stocks for a commodity serves as a negative signal for futures prices. This is due to the fact that higher ending stocks means that there is an excess of supply over expectations, which in turn puts downward pressure on futures prices [2]. After the WASDE report is released, the participants in the commodity markets obtain new information about the predicted ending stocks and adjust their trading behaviors, which will move the futures price in the direction depending on whether the surprise of ending

[1] The daily futures basis is constructed as $CAR_{i,t} = log(\frac{F_{i,t}^{(1)}}{F_{i,t}^{(0)}})$, where $F_{i,t}^{(0)}$ and $F_{i,t}^{(1)}$ are front-month futures price and the next-maturity futures price on day t of commodity i. Following Amihud (2002), the daily illiquidity measure is calculated as $AMI_{i,t} = \frac{|R_{i,t}|}{FVOL_{i,t}}$, where $|R_{i,t}|$ and $FVOL_{i,t}$ are the absolute value of daily futures return and futures volume on the day t for commodity i.

[2] See Appendix A for details

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stock forecasts is positive or negative. Thus, the negative relation between the change in forecasts of ending stocks and post-announcement returns is not surprising. However, when the cumulative return horizons extend beyond CUM(+0, +5) to include CUM(+0, +10), CUM(+0, +15), and CUM(+0, +20), the predictive power of ΔES diminishes for cumulative post-announcement returns. This suggests that ending stocks forecasts from the WASDE reports do not demonstrate reliable predictive capabilities over longer time periods. Following the incorporation of information from ending stocks forecasts into futures prices, the predictive efficacy of ΔES diminishes rapidly.

The coefficient of options volumes is strongly negative (t-statistics = -2.888) in the column of CUM(+0,+5). When informed traders in options markets obtain private information before the announcement, they will make decisions about investment before the report becomes publicly available, which leads to fluctuations in trading volumes. Furthermore, options volumes have significant predictive power (t-statistics = -2.786, -3.502, -3.737) for the other three columns of CUM(+0,+10), CUM(+0,+15), and CUM(+0,+20). An important finding is that options volumes even have better performance when the predictive horizon is longer since t-statistics are larger in CUM(+0,+15) and CUM(+0,+20) than CUM(+0,+5). Futures volumes also have predictive power for futures returns in post-announcement periods (t-statistics = 2.136, 1.792, 2.499, 2.433) in the four return windows within twenty days after the release date. The trading volume of options and futures in pre-announcement (day -5 to day -1) not only contains the information regarding the WASDE reports but also includes news that is farther away than the post-announcement window (day +5 to day +20). Thus, trading volumes in commodity markets have a longer horizon in predictive power than the change of predictions in ending stocks ΔES. Consistent with the findings presented in the previous section that more informed trading happens in options than futures, the information contained in options markets during the pre-announcement period exhibits superior predictive capabilities compared to that in futures markets. The variable pre-CUM is positive and significant (t-statistics = 2.142, 1.668, 1.731) in the columns of CUM(+0,+5),

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CUM(+0,+10), and CUM(+0,+15), which indicates the momentum effects exist in before and after the announcement. However, the momentum effect diminishes as the time horizon extends over a longer period.

5 Conclusion

In this study, we examine the USDA announcement effects on commodity markets. In the previous literature, most related studies focus on the reactions in commodity futures markets from WASDE reports, whether commodity options markets are affected has received much less attention. However, as a critical part of commodity markets, it is of great importance to explore how commodity options markets react to WASDE reports. We analyze how the abnormal futures and options volumes change before and after the released date and conclude that the USDA announcement effects exist in both futures and options markets. Specifically, the effects are more significant in futures than options. In most commodities, these effects fade away within five days in most futures and options markets, while the decline speed is slower in options. Furthermore, our study provides evidence that options markets are involved with more informed trading behaviors than futures since there are more options markets that exist significant abnormal options volumes during the pre-announcement period. Motivated by these findings, we conduct the analysis of post-announcement returns and compare the predictive power between the information in commodity markets and ending stocks forecasts from WASDE reports. The results show that commodity futures and options have a longer predictive horizon than ending stocks forecasts. Our work should provide guidance for further work in the area of commodity options markets and WASDE reports. For example, a detailed analysis to explore the reason why the signs of coefficients for futures and options volumes in predicting post-announcement returns are opposite is an interesting avenue for future work.

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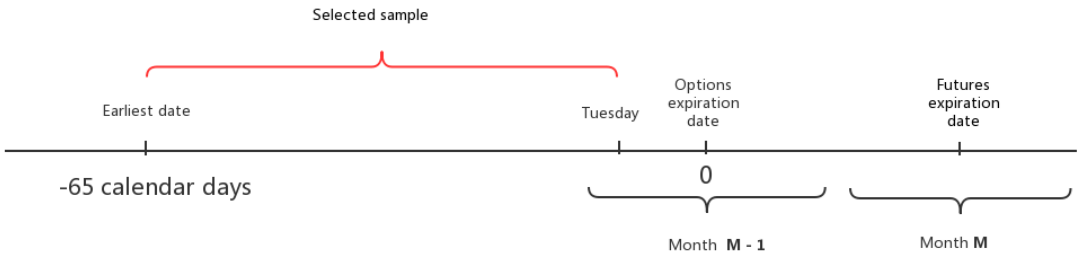


Figure 1: Procedure to select sample (source: Zhang (2022))

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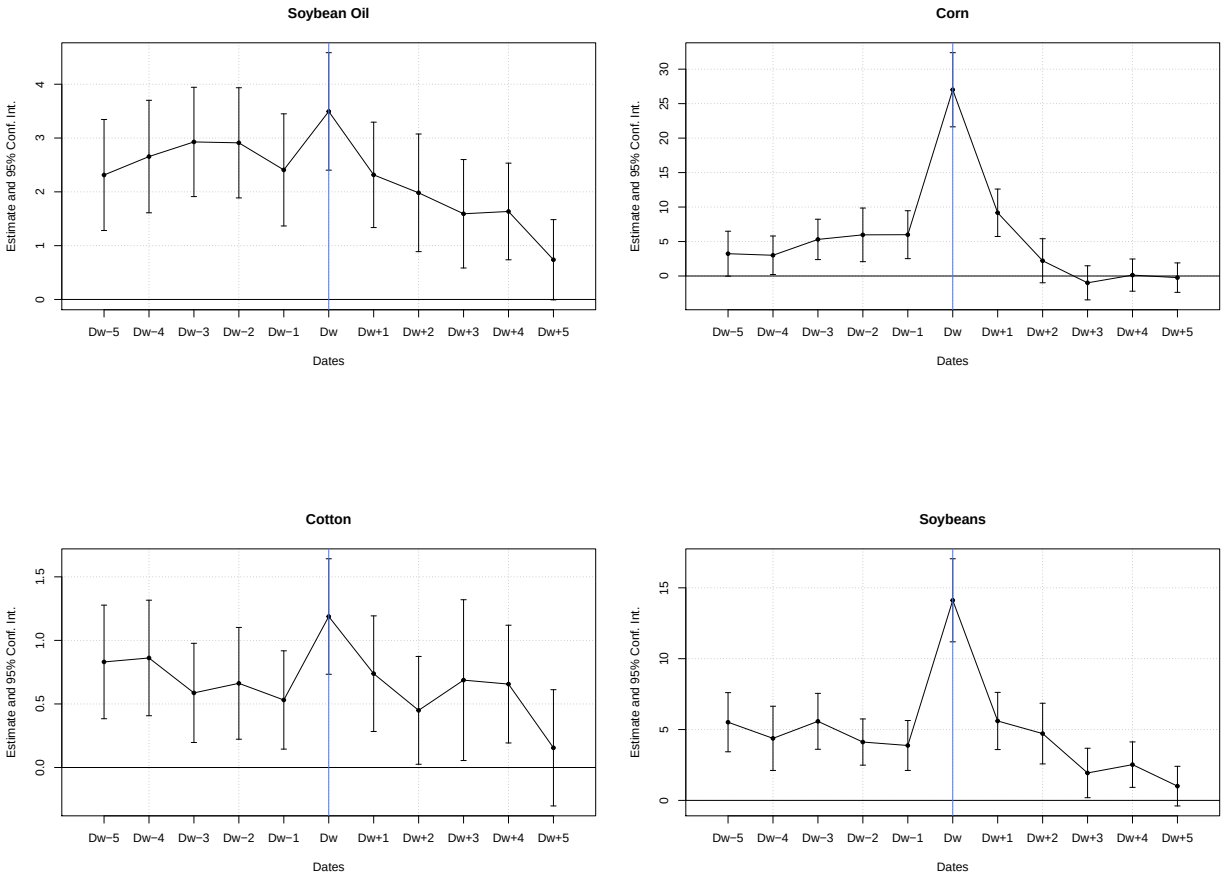


Figure 2: The coefficient and 95% confidence interval of abnormal futures volumes around announcement date (1)

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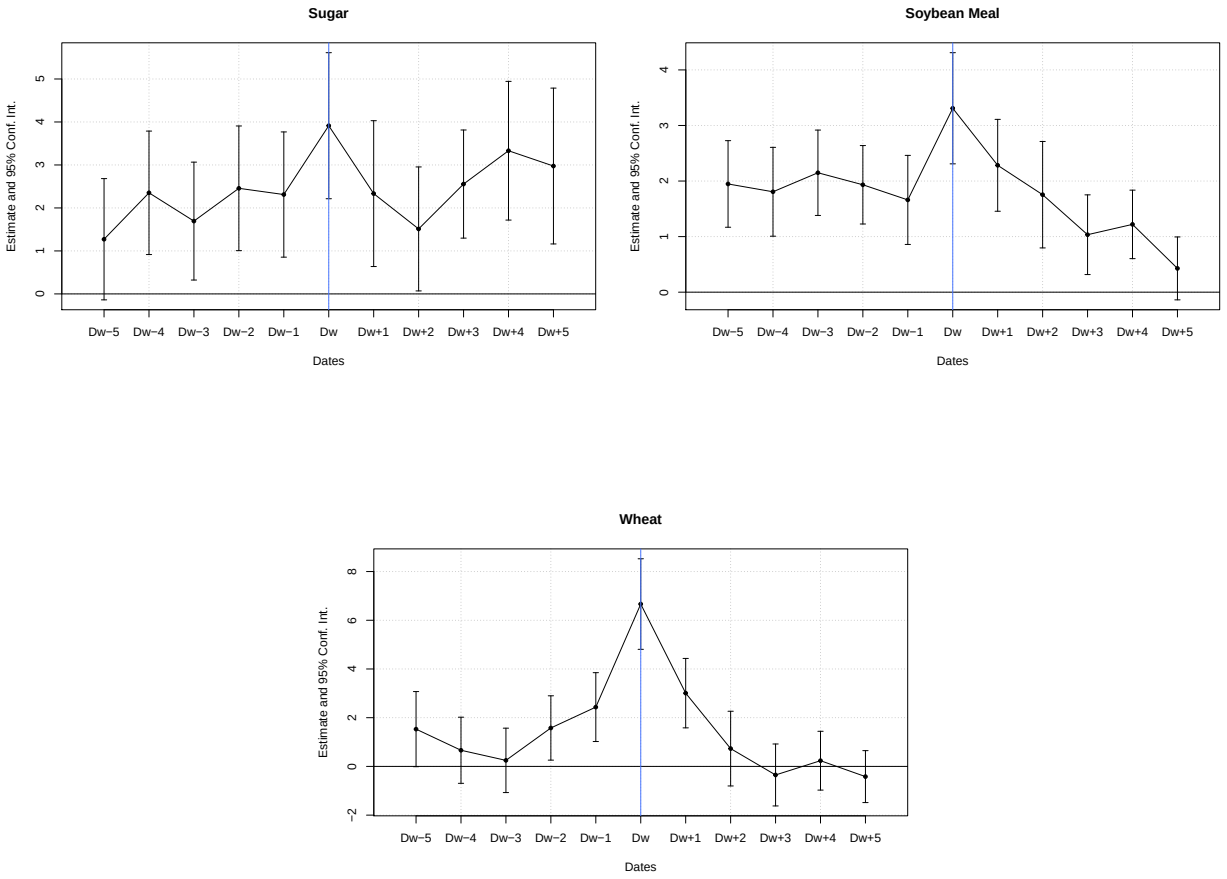


Figure 3: The coefficient and 95% confidence interval of abnormal futures volumes around announcement date (2)

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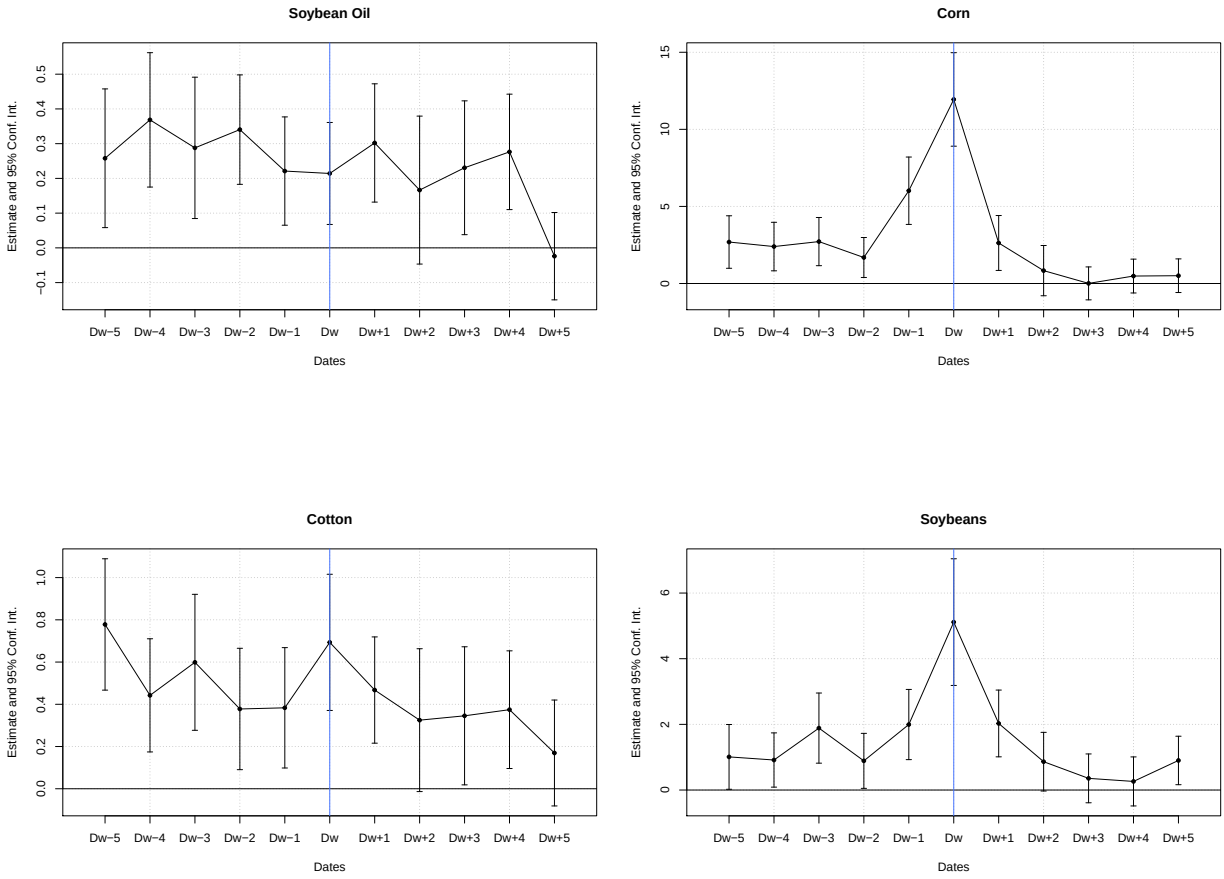


Figure 4: The coefficient and 95% confidence interval of abnormal options volumes around announcement date (1)

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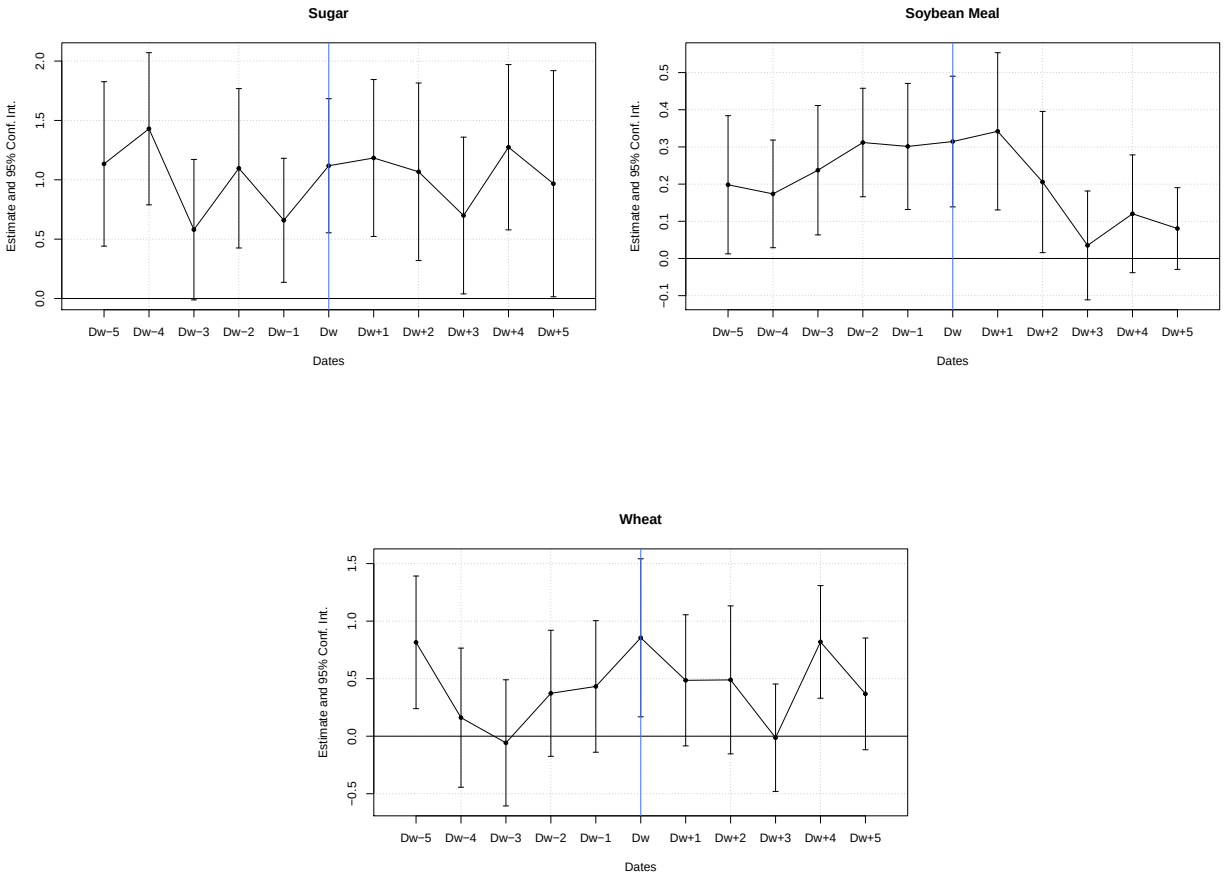


Figure 5: The coefficient and 95% confidence interval of abnormal options volumes around announcement date (2)

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Table 1: Summary statistics for commodities in the sample.

This table shows the summary statistics of seven agricultural commodity futures and options. Columns $FVOL$ and $OVOL$ shows the average daily futures and options trading volumes. Columns E[R](%) and $\sigma[R]$, are the average and standard deviation, respectively, of daily futures returns.

Commodity	Symbol	$FVOL$ ($\times 10^3$)	$OVOL$ ($\times 10^3$)	E[R](%)	SD[R]	Sharpe ratio
Wheat	W	37.962	9.552	-0.052	1.875	-0.028
Corn	C	93.736	27.781	-0.030	1.652	-0.018
Soybeans	S	58.429	19.102	0.007	1.448	0.005
Soybean oil	BO	25.287	2.216	-0.018	1.422	-0.013
Soybean meal	SM	21.490	2.171	0.039	1.605	0.024
Sugar	SB	36.997	8.296	-0.049	2.001	-0.024
Cotton	CT	9.515	3.460	-0.027	1.602	-0.017

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Table 2: Abnormal values of futures and commodity volumes

This table presents the abnormal values of trading volumes in commodity markets. Columns ab_FVOL ($\times 10^3$) and ab_OVOL ($\times 10^3$) show the abnormal values of futures and options volumes. DW denotes the announcement date.

	Wheat		Corn		Soybeans		Soybean oil		Soybean meal		Sugar		Cotton	
Dates	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL	ab_FVOL	ab_OVOL
Dw-5	40.304	10.584	95.712	27.400	61.294	19.625	25.886	2.431	21.937	2.232	37.504	8.356	9.819	3.565
Dw-4	38.774	9.384	92.018	25.532	59.161	18.087	24.476	2.210	20.730	2.005	36.707	7.389	9.189	3.072
Dw-3	38.857	8.980	95.132	26.361	59.277	17.883	24.892	2.164	20.993	2.117	38.314	7.511	9.627	3.383
Dw-2	40.839	9.087	98.155	26.351	58.559	17.539	25.636	2.109	21.363	2.149	40.330	7.519	9.961	3.448
Dw-1	42.260	10.007	101.197	33.949	58.484	20.666	25.967	2.174	21.687	2.447	40.878	6.942	9.931	3.576
Dw	49.931	10.998	129.645	42.023	76.552	26.756	29.847	2.497	26.584	2.642	42.757	7.550	11.623	4.009
Dw+1	42.367	9.338	107.192	30.268	62.732	20.190	26.215	2.122	22.340	2.447	41.006	8.162	10.145	3.710
Dw+2	39.500	8.986	98.632	25.957	61.223	18.827	25.900	2.074	22.368	2.214	36.198	9.041	9.540	3.628
Dw+3	37.179	9.088	93.592	25.719	57.677	17.697	25.086	2.168	21.393	2.133	34.988	8.667	9.694	3.517
Dw+4	34.056	8.846	87.793	24.614	55.528	17.820	24.086	2.058	20.584	1.965	37.482	9.965	9.588	3.628
Dw+5	34.503	9.507	83.761	24.820	54.421	17.528	23.168	1.930	20.087	1.983	36.871	10.273	9.490	3.587

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Table 3: Results of USDA announcement effect in futures volumes (1)

This table presents the analysis of abnormal values of futures volumes using the ordinary least squares regression. Seasonal represents the monthly dummy variables. The time trend increases by one unit per calendar year. Control variables are the futures volumes and return in the previous day of the announcement date. The standard errors are estimated by Newey-West method to control serial correlation. The t-statistics are shown in parentheses. The notations ***, **, * indicate the coefficient is significant at the 1%, 5%, and 10% level, respectively.

Dependent variable: ab_FVOL								
	Soybean Oil		Corn		Cotton		Soybean	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dw-5	2.020***	2.313***	1.816	3.235*	0.933***	0.831***	4.993***	5.517***
	(3.637)	(4.395)	(1.098)	(1.945)	(3.889)	(3.643)	(4.404)	(5.180)
Dw-4	2.418***	2.656***	2.071	3.007**	0.929***	0.862***	3.910***	4.378***
	(4.254)	(4.980)	(1.491)	(2.108)	(3.775)	(3.716)	(3.184)	(3.787)
Dw-3	2.759***	2.928***	4.357***	5.308***	0.580***	0.587***	4.992***	5.580***
	(5.067)	(5.649)	(2.954)	(3.555)	(2.769)	(2.948)	(4.679)	(5.545)
Dw-2	2.987***	2.911***	5.346***	5.969***	0.698***	0.662***	3.625***	4.117***
	(5.328)	(5.573)	(2.695)	(3.009)	(2.906)	(2.952)	(4.001)	(4.952)
Dw-1	2.439***	2.408***	5.014***	5.992***	0.447**	0.531***	3.196***	3.875***
	(4.345)	(4.529)	(2.831)	(3.380)	(2.151)	(2.693)	(3.387)	(4.312)
Dw	3.420***	3.494***	25.875***	27.013***	1.136***	1.188***	13.778***	14.119***
	(5.960)	(6.274)	(9.438)	(9.849)	(4.623)	(5.124)	(9.054)	(9.442)
Dw+1	2.195***	2.316***	7.716***	9.171***	0.714***	0.738***	5.145***	5.605***
	(4.219)	(4.634)	(4.381)	(5.227)	(3.042)	(3.182)	(4.792)	(5.443)
Dw+2	1.425**	1.982***	−1.125	2.213	0.257	0.450**	3.166***	4.715***
	(2.541)	(3.554)	(−0.673)	(1.357)	(1.177)	(2.078)	(2.850)	(4.316)
Dw+3	1.464***	1.593***	−2.683**	−0.989	0.620*	0.687**	1.229	1.935**
	(2.721)	(3.098)	(−2.091)	(−0.785)	(1.934)	(2.130)	(1.337)	(2.173)
Dw+4	1.584***	1.635***	−0.882	0.131	0.597**	0.656***	1.956**	2.523***
	(3.255)	(3.569)	(−0.715)	(0.110)	(2.440)	(2.777)	(2.293)	(3.084)
Dw+5	0.772*	0.738*	−0.771	−0.236	0.090	0.155	0.679	1.007
	(1.838)	(1.941)	(−0.712)	(−0.217)	(0.382)	(0.664)	(0.910)	(1.411)
Time Trend	No	Yes	No	Yes	No	Yes	No	Yes
Seasonal	No	Yes	No	Yes	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,722	5,722	4,834	4,834	4,529	4,529	5,657	5,657
R^2	0.164	0.335	0.243	0.275	0.144	0.212	0.152	0.279

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Table 4: Results of USDA announcement effect in futures volumes (2)

This table presents the analysis of abnormal values of futures volumes using the ordinary least squares regression. Seasonal represents the monthly dummy variables. The time trend increases by one unit per calendar year. Control variables are the futures volumes and return in the previous day of the announcement date. The standard errors are estimated by Newey-West method to control serial correlation. The t-statistics are shown in parentheses. The notations ***, **, * indicate the coefficient is significant at the 1%, 5%, and 10% level, respectively.

Dependent variable: ab_FVOL						
	Sugar		Soybean Meal		Wheat	
	(1)	(2)	(3)	(4)	(5)	(6)
Dw-5	1.378*	1.272*	1.866***	1.948***	1.499*	1.531*
	(1.948)	(1.767)	(4.283)	(4.899)	(1.894)	(1.946)
Dw-4	2.344***	2.352***	1.744***	1.807***	0.730	0.663
	(3.212)	(3.210)	(3.804)	(4.429)	(1.039)	(0.958)
Dw-3	1.765***	1.694**	2.074***	2.149***	0.362	0.248
	(2.595)	(2.421)	(4.567)	(5.483)	(0.513)	(0.368)
Dw-2	2.473***	2.457***	2.001***	1.933***	1.778**	1.578**
	(3.197)	(3.320)	(4.588)	(5.366)	(2.565)	(2.342)
Dw-1	2.226***	2.312***	1.672***	1.660***	2.599***	2.435***
	(2.854)	(3.110)	(3.582)	(4.060)	(3.395)	(3.376)
Dw	3.708***	3.913***	3.317***	3.310***	6.687***	6.665***
	(4.204)	(4.515)	(6.181)	(6.485)	(6.826)	(7.027)
Dw+1	2.079**	2.333***	2.262***	2.283***	2.921***	3.008***
	(2.432)	(2.696)	(4.589)	(5.412)	(4.003)	(4.136)
Dw+2	1.187	1.512**	1.327**	1.754***	0.121	0.731
	(1.575)	(2.055)	(2.575)	(3.589)	(0.163)	(0.934)
Dw+3	2.356***	2.556***	0.917**	1.034***	−0.470	−0.351
	(3.490)	(3.983)	(2.128)	(2.826)	(−0.712)	(−0.541)
Dw+4	3.186***	3.331***	1.148***	1.220***	0.279	0.235
	(3.734)	(4.046)	(2.924)	(3.879)	(0.448)	(0.381)
Dw+5	2.931***	2.975***	0.419	0.428	−0.252	−0.419
	(3.162)	(3.216)	(1.261)	(1.481)	(−0.461)	(−0.769)
Time Trend	No	Yes	No	Yes	No	Yes
Seasonal	No	Yes	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,837	3,837	5,708	5,708	4,947	4,947
R^2	0.189	0.206	0.185	0.319	0.257	0.287

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Table 5: Results of USDA announcement effect in option volumes (1)

This table presents the analysis of abnormal values of option volumes using the ordinary least squares regression. Seasonal represents the monthly dummy variables. The time trend increases by one unit per calendar year. Controls variables are the options volumes and return in the previous day of the announcement date. The standard errors are estimated by Newey-West method to control serial correlation. The t-statistics are shown in parentheses. The notations ***, **, * indicate the coefficient is significant at the 1%, 5%, and 10% level, respectively.

Dependent variable: ab_OVOL								
	Soybean Oil		Corn		Cotton		Soybean	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dw-5	0.221**	0.258**	1.900**	2.691***	0.832***	0.778***	0.731	1.009**
	(2.001)	(2.532)	(2.140)	(3.097)	(4.927)	(4.903)	(1.351)	(2.005)
Dw-4	0.335***	0.368***	1.872**	2.397***	0.495***	0.442***	0.797*	0.914**
	(3.162)	(3.734)	(2.316)	(2.985)	(3.320)	(3.235)	(1.826)	(2.169)
Dw-3	0.253**	0.288***	2.263***	2.719***	0.604***	0.599***	1.695***	1.886***
	(2.400)	(2.773)	(2.868)	(3.410)	(3.645)	(3.645)	(3.075)	(3.462)
Dw-2	0.333***	0.340***	1.414**	1.687**	0.408***	0.378***	0.853*	0.888**
	(3.969)	(4.236)	(2.156)	(2.551)	(2.728)	(2.576)	(1.915)	(2.078)
Dw-1	0.220**	0.221***	5.577***	6.018***	0.355**	0.383***	1.921***	1.995***
	(2.548)	(2.780)	(5.028)	(5.398)	(2.412)	(2.636)	(3.438)	(3.657)
Dw	0.216***	0.214***	11.545***	11.940***	0.682***	0.693***	5.183***	5.115***
	(2.779)	(2.860)	(7.364)	(7.717)	(4.129)	(4.213)	(5.344)	(5.196)
Dw+1	0.302***	0.302***	1.522*	2.629***	0.461***	0.467***	1.749***	2.028***
	(3.255)	(3.477)	(1.720)	(2.894)	(3.507)	(3.641)	(3.315)	(3.908)
Dw+2	0.134	0.166	−0.907	0.835	0.283	0.325*	−0.006	0.863*
	(1.247)	(1.531)	(−1.070)	(1.004)	(1.590)	(1.882)	(−0.014)	(1.893)
Dw+3	0.234**	0.231**	−0.706	0.008	0.335**	0.345**	0.088	0.355
	(2.321)	(2.346)	(−1.162)	(0.014)	(1.999)	(2.070)	(0.215)	(0.936)
Dw+4	0.290***	0.276***	0.203	0.484	0.372**	0.374***	0.152	0.262
	(3.246)	(3.259)	(0.329)	(0.864)	(2.558)	(2.634)	(0.364)	(0.687)
Dw+5	−0.027	−0.024	0.419	0.506	0.171	0.169	0.893**	0.901**
	(−0.387)	(−0.373)	(0.722)	(0.909)	(1.324)	(1.324)	(2.295)	(2.390)
Time Trend	No	Yes	No	Yes	No	Yes	No	Yes
Seasonal	No	Yes	No	Yes	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,722	5,722	4,834	4,834	4,529	4,529	5,657	5,657
R^2	0.147	0.228	0.225	0.277	0.152	0.217	0.192	0.264

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Table 6: Results of USDA announcement effect in option volumes (2)

This table presents the analysis of abnormal values of option volumes using the ordinary least squares regression. Seasonal represents the monthly dummy variables. The time trend increases by one unit per calendar year. Controls variables are the options volumes and return in the previous day of the announcement date. The standard errors are estimated by Newey-West method to control serial correlation. The t-statistics are shown in parentheses. The notations ***, **, * indicate the coefficient is significant at the 1%, 5%, and 10% level, respectively.

Dependent variable: ab_OVOL						
	Sugar		Soybean Meal		Wheat	
	(1)	(2)	(3)	(4)	(5)	(6)
Dw-5	0.963***	1.134***	0.184*	0.198**	0.626**	0.816***
	(2.733)	(3.209)	(1.812)	(2.092)	(2.010)	(2.774)
Dw-4	1.258***	1.430***	0.170**	0.174**	−0.072	0.161
	(3.739)	(4.373)	(2.232)	(2.351)	(−0.235)	(0.522)
Dw-3	0.428	0.580*	0.229**	0.237***	−0.263	−0.058
	(1.494)	(1.922)	(2.472)	(2.674)	(−0.925)	(−0.207)
Dw-2	0.981***	1.097***	0.319***	0.312***	0.283	0.373
	(3.253)	(3.203)	(4.411)	(4.189)	(1.087)	(1.334)
Dw-1	0.603**	0.659**	0.299***	0.301***	0.354	0.432
	(2.569)	(2.472)	(3.400)	(3.484)	(1.247)	(1.482)
Dw	1.124***	1.118***	0.315***	0.315***	0.762**	0.855**
	(3.849)	(3.881)	(3.717)	(3.507)	(2.220)	(2.440)
Dw+1	1.332***	1.184***	0.325***	0.342***	0.301	0.486*
	(3.673)	(3.511)	(3.171)	(3.171)	(1.107)	(1.671)
Dw+2	1.171***	1.068***	0.179*	0.206**	0.216	0.489
	(3.273)	(2.797)	(1.903)	(2.121)	(0.664)	(1.491)
Dw+3	0.792**	0.699**	0.014	0.035	−0.126	−0.013
	(2.360)	(2.075)	(0.194)	(0.470)	(−0.510)	(−0.056)
Dw+4	1.315***	1.274***	0.116	0.120	0.746***	0.820***
	(3.199)	(3.588)	(1.577)	(1.487)	(2.961)	(3.283)
Dw+5	1.012**	0.967**	0.078	0.081	0.317	0.368
	(2.193)	(1.990)	(1.469)	(1.437)	(1.305)	(1.484)
Time Trend	No	Yes	No	Yes	No	Yes
Seasonal	No	Yes	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,837	3,837	5,708	5,708	4,947	4,947
R^2	0.148	0.199	0.206	0.248	0.207	0.252

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Table 7: Results for post-announcement cumulative returns analysis

This table presents the ordinary least squares regression (the standard errors are clustered by time) results of the predictive power of futures and options volumes on the cumulative returns following the announcement. ΔES is the change of forecast in ending stock from last month to current month, scaled by the forecast in the last month. pre-CUM denotes cumulative return over the pre-announcement window (days -5 to -1); abs(pre-CUM) denotes absolute value ofcumulative return over the pre-announcement window (days -5 to -1). CAR and AMI are the average value of daily futures basis and measure for illiquidity for commodity over the pre-announcement window. The t-statistics are shown in parentheses. The notations ***, **, * indicate the coefficient is significant at the 1%, 5%, and 10% level, respectively.

Dep. variable:	CUM(+0,+5)	CUM(+0,+10)	CUM(+0,+15)	CUM(+0,+20)
ΔES	-3.800^{***}	0.132	0.045	0.022
	(-3.759)	(1.171)	(0.463)	(0.270)
log(F_VOL)	0.285^{**}	0.034^{*}	0.043^{**}	0.037^{**}
	(2.136)	(1.792)	(2.499)	(2.433)
log(O_VOL)	-0.275^{***}	-0.039^{***}	-0.044^{***}	-0.040^{***}
	(-2.888)	(-2.786)	(-3.502)	(-3.737)
pre-CUM	0.080^{**}	0.008^{*}	0.009^{*}	0.007
	(2.142)	(1.668)	(1.731)	(1.592)
abs(pre-CUM)	-0.064	-0.007	-0.002	-0.003
	(-1.254)	(-0.961)	(-0.225)	(-0.676)
CAR	7.052^{*}	0.695	0.973^{**}	0.565
	(1.952)	(1.474)	(2.130)	(1.579)
AMI	-129.378	67.304	70.972	17.684
	(-0.225)	(0.809)	(1.134)	(0.244)
Constant	-7.641^{**}	-0.773	-1.074^{**}	-0.637^{*}
	(-2.046)	(-1.581)	(-2.287)	(-1.744)
Observations	2,173	2,074	2,074	2,074
R^2	0.023	0.011	0.016	0.013

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Appendices

A Model for ending stocks and futures prices

Following the rational storage model in Zhang and Li (2022), the amount of crop to be stored to the following period shall be compensated by the carry in the futures markets:

$$E_t[p_T(z_T + c_T)] - p_t = g(c_T|c_t)$$

where p_T, c_T, and z_T are the commodity price, the level of ending stocks, and the new production of the crop at the end of marketing year T. The spot price and the carryout of commodity from the previous marketing year are p_t and c_t. What's more, $g(c_T|c_t)$ is the cost function of storing c_T, conditional on the initial carryout c_t.

If we assume that there is the absence of a risk premium, the expected commodity prices equal to futures prices with the same maturity date T:

$$E_t[p_T(z_T + c_T)] = f_{t,T}$$

since the price function p_T is decreasing in c_T, there is a negative relationship between c_T and $f_{t,T}$.

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